ULTRA HDI DESIGN GUIDELINES 2.0

NCAB GROUP

Ultra HDI PCB is defined as a PCB with attributes beyond the existing IPC-2226 A producibility level C, where one or more layers have lines or spaces below 50um. This is the current IPC definition of an Ultra HDI PCB, but we can also mention typical characteristics such as:

- Microvia diameters below 75 micron.
- Wire bonding pads of 125um or less,
- Wire bonding pitch below 150um,
- BGA pitch below 400um.
- Dielectric insulation between layers below 50um
- Pad size tolerances +/-10% or better

This guideline do not refer to a reliability level such as IPC class 2 or 3 since IPC standards such as IPC-2226 or IPC-6010 series are limited to lines and spaces of 50um or more. Beyond this level any acceptability measures are As Agreed Between User and Supplier (AABUS).

This document shall be used as a guideline and will be **subject to continuous review and improvement**. The parameters given are compromises to fit more than one factory. Still, combinations of parameters close to the limit could be outside our capability window.

We always request our customers to check with their local NCAB FAE before completing a design.

NCAB design steps related to production technologies

We classify our Factories in accordance to these steps. The reason is we recognize the need for different production technologies and equipment changes as we go down the ladder. This can be seen in several tables, still some parameters are common for several steps in this ladder.

Line and Space (um)	PCB Technology
50-40	Traditional HDI manufactuiring in an advanced factory
40-30	High end HDI manufacturing, but also m-SAP
30-20	mSAP, and Embedded Trace
20-10	Advanced SAP, Embedded Trace and IC Substrate level technologies
< 10	Not NCAB focus area (until further notice)

Technology

MAIN PRODUCTION PROCESS	TRACK / GAP RANGES	REMARK
mSAP set up	> 20/20um track / gap	Other processes can be a limitation
Tenting process	>=30/30um track / gap	Require updated cleanrooms and other equipment as well.

TYPICAL BOARD THICKNESS	LAYER COUNT RANGE	REMARK
Board thickness <=0.5mm	1-6 layers	IC Substrate – thickness limited by mSAP process lines – for mSAP cores and/or outer layers.
Board thickness <=1.2mm	2-6 layers	Traditional IC substrate, 8-12 layers on request
Board thickness 1.2-2.0mm	8-16 layers	Substrate like PCB – SPL
Board thickness >2.0mm (max 8mm)	>16 layers (max > 100 layers)	Test boards like ATE, BIB and with DUT areas

Available materials (depending on factory and technology)

SUPPLIERS	MGC	HITACHI	PANASONIC	SHENGYI	DOOSAN	OTHERS
Type of CCL materials	832 NXA/NS – HL972LF/LD	679FGB/S700G/705G/770G/ LW900G/LW910G	R-1515E	SI13U/SI10U	7409HGB series	ABF

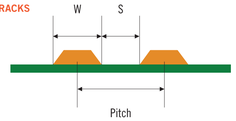
SUPPLIERS	TAIYO	OTC	HITACHI	SUMITOMO
Type of Solder Mask	PSR-4000 AUS 308/320(Green) PSR-800 AUS SR1 / AUS 410(Green, D/F) PSR4000 AUS MEH/EG23(Black)	Z23(Green)	FZ2700G	LAZ-7751/52

Design tips

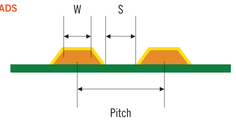
1. Reduce track before space when smaller size is needed.
2. Every stackup must be approved case by case, specially when coreless or odd layers are requested..
3. Minimum dielectric thickness of core and prepreg can be 0.025mm (1mil).
4. Overall thickness determines manufacturing method and factory: IC substrates < 0.8mm, some factories < 0.5mm. Subtractive method < 2mm. Test boards (ATE/BIB etc) 8-10mm.
5. Avoid Plated Through Holes when start/stop layers are UHDI
6. Copper thickness in UHDI layers will be thinner than the normal HDI layers. Exact start copper and final copper must be evaluated in each case. Normally, the minimum final copper thickness of conductors will be 8-30um.
7. Cores with "X" micro-via drilling is available on request. Core thickness between 60-160mm with hole size<=80um. See table "X-Via".
8. For bonding pads, the width will be measured on the top of the pad. Required minimum bonding pad area must be specified and agreed.
9. Check with local office if the tolerance of pad size <=+/-10um needed. The advanced capability now is +/-5um.
10. For SM flatness 5-7um, dry film solder mask will be used, but availability is limited to Substrate or Substrate Like PCB factories.
11. All designs with Track/ gap less than 35um must be evaluated case by case, and typically with limited layer count.
12. Surface treatment ENIG, ENEPIG, OSP, Immersion Tin and soft gold plating is available. But for FCBGA, ENEPIG is recommended but less available. Immersion tin is recommended in some factories.

Design rules

MEASUREMENT OF TRACKS



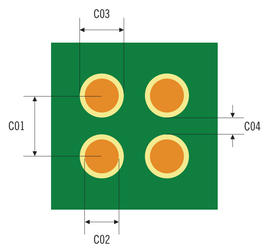
MEASUREMENT OF PADS



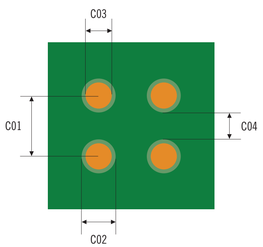
PITCH OF PADS OR TRACKS (W/S UM)

BASE COPPER	PAD PITCH						
	100	80	70	60	50	40	30
0	50/50	40/40	35/35	30/30	25/25	20/20	15/15
3	50/50	40/40	35/35	30/30	25/25	20/20	15/15
6	50/50	40/40	35/35	30/30	25/25	20/20	/
9	50/50	40/40	35/35	30/30	25/25	/	/
12	50/50	40/40	35/35	30/30	/	/	/
15	50/50	40/40	/	/	/	/	/
18	50/50	/	/	/	/	/	/

NON SOLDERMASK DEFINED



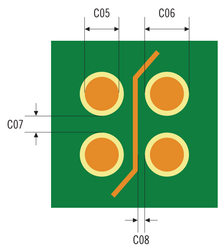
SOLDERMASK DEFINED (WHEN C03<C02)



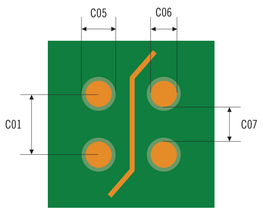
VIA IN PADS PITCH (UM) WITHOUT TRACKS BETWEEN PADS

ITEMS	VIA IN PADS PITCH (UM) WITHOUT TRACKS BETWEEN PADS					
Pitch (C01)	400	350	300	250	200	150
NSMD Pad (C02)	250	200	200	150	110	80
Pad Tolerance	+/-20	+/-15	+/-15	+/-10	+/-10	+/-10
Pad to Pad spacing	150	150	100	100	90	70
NSMD SM Clearance	50	50	30	30	30	30
NSMD SM Opening (C03)-**ENIG**	350	300	260	210	170	140
NSMD SM Opening (C03)-**Au plating**	340	290	250	200	160	130
NSMD SM Opening (C03)-**OSP**	330	280	240	190	150	120
SM Opening tolerance	+/-25	+/-20	+/-20	+/-20	+/-20	/
SM dam. (C04)	50	50	40	40	30	/
Micro-via size	100	80	80	60	60	/

NON SOLDERMASK DEFINED



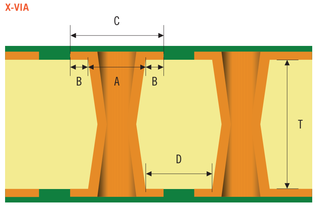
SOLDERMASK DEFINED (WHEN C06<C05)



VIA IN PADS PITCH (UM) WITH 1 TRACK BETWEEN PADS

ITEMS	VIA IN PADS PITCH (UM) WITH 1 TRACK BETWEEN PADS					
Pitch	400	350	300	250	225	180
NSMD Pad (C05)	200	180	150	150	150	120
Pad Tolerance	+/-20	+/-15	+/-10	+/-10	+/-10	+/-10
NSMD SM Clearance	40	35	30	/	/	/
NSMD SM Opening (C06)-**ENIG**	280	250	210	100	100	60
NSMD SM Opening (C06)-**Au plating**	270	240	200	100	100	60
NSMD SM Opening (C06)-**OSP**	270	240	200	100	100	60
SM Opening tolerance	+/-25	+/-20	+/-20	+/-20	+/-20	+/-20
SM dam. (C07)	120	100	90	150	125	120
SM Coating on track (C08)	35	35	35	/	/	/
Micro-via size	75	70	60	60	60	60
Track width	50	30	30	33.3	25	20
Track spacing	75	70	60	33.3	25	20
Conductor thickness	18	18	18	12	9	9

X-VIA



X-VIA RULE

NO.	ITEM	HVM (UM)	NPI (UM)
A	Via diameter	Min. 50	Min. 45
B	Annular ring	Min. 30	Min. 25
C	Via land	A+B*2	A+B*2
D	Space between holes (Same net)	Min. 75	Min. 60
	Space between holes (Different net)	Trace to trace space @ Finished copper + B*2	Trace to trace space @ Finished copper + B*2
T	Core thickness	Max. 160	Max. 200

Tolerance (A): ±15

X-VIA HOLE RECOMMENDED DESIGN (UM)

VIA SIZE (UM)	50-60	50	60	70	80
Core Thickness (Max. mm)	<=0.08	0.1	0.12	0.14	0.16

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